

SRM Institute of Science and Technology

Course Code: 21CSC302J | Course Name: COMPUTER NETWORKS

Assignment 5

Unit-5: Transport and Application Layer Protocols

Instructions: Answer all questions. Draw neat diagrams wherever necessary. Show complete working for numerical questions.

- Q1.** Explain the role of port numbers in transport layer communication, and give examples of well-known port numbers.
- Q2.** Compare the User Datagram Protocol (UDP) and Transmission Control Protocol (TCP) in terms of reliability, connection type and header size.
- Q3.** Explain the TCP segment header format, describing the function of at least six important fields.
- Q4.** Explain the TCP three-way handshake process used to establish a connection, with a neat diagram.
- Q5.** Explain the World Wide Web (WWW) and the HTTP protocol, including the request-response model.
- Q6.** Explain the File Transfer Protocol (FTP), differentiating between active mode and passive mode.
- Q7.** Explain the email architecture and the roles of SMTP, POP3 and IMAP protocols.
- Q8.** Explain the Telnet protocol, its working and typical use cases.
- Q9.** Explain the Domain Name System (DNS) and describe its hierarchical namespace structure.
- Q10.** Explain, step by step, how a DNS query is resolved from a client requesting `www.example.com` to receiving the corresponding IP address.

--- End of Assignment ---